unsolved problems in certain determinate laws of nature, such as Coulomb's law of electrical attraction; for he cannot be satisfied with Coulomb's formula for the force or the potential, but continues to search for exceptions, and is not content until he succeeds in determining the magnitude of the probability that the electrical force differs from Coulomb's value by some arbitrary pre-assigned value.

The determinist thinks along the very opposite lines in all these points. He assigns to Coulomb's law the satisfactory character of absolute validity. On the other hand, he recognizes the wave function as a mere probability magnitude only so long as the measuring instrument used in the study of the wave is disregarded, and he looks for inflexible laws connecting the properties of the wave function and the occurrences in the measuring apparatus. For this purpose, he must obviously include in the subject matter of his investigation both the wave function and the measuring instruments. In other words, he must translate into his physical world picture not only the entire experimental setup used for the production of the material waves—such as the high voltage battery, the incandescent filament, the radioactive compound—but also the measuring apparatus, such as the photographic plate, the ionization chamber, the point counter, together with all the processes occurring therein, and he must deal with all these objects as one single structure, as an isolated system. But this does not yet take care of the problem. On the contrary—it makes it even more complicated. For in this case, if the total structure is to retain its unique character, it must neither be split up nor exposed to external influences, so that a direct examination is completely impossible. Nevertheless it now becomes possible to formulate certain new hypotheses concerning the internal processes of the system, and subsequently to test their consignments. Whether this procedure leads to any actual advance, is a question which only the future can answer. For the time being, it is still impossible to determine clearly what direction will lead to progress. At any rate, all the circumstances mentioned prove that the elementary quantum of action erects an objective barrier which limits the efficiency of the measuring apparatus available to physical science, and that, therefore, the desired progress will only give this barrier even sharper outlines than it had before.

Properly speaking, we have thus reached the end of our considerations which have demonstrated that an adherence to a strictly causal outlook—always taking the word "causal" in the modified sense explained previously—is by no means excluded from the viewpoint of even modern physics, though its necessity can never be proved either *a priori or a posteriori.* Nevertheless, not even a convinced determinist—indeed, especially not a convinced determinist—can escape at this point some doubts which prevent him from